Brain Tumor Classification from MRI:

A Label-Efficient and Leakage-Free Deep Learning Approach

Research Concept Note — prepared for review by the project supervisor

Dataset: 4-class brain MRI (glioma, meningioma, no-tumor, pituitary)

1. Objective of the Study

The aim of this project is to build a computer model that can look at a brain MRI scan and automatically decide which of four categories it belongs to: a **glioma** tumor, a **meningioma** tumor, a **pituitary** tumor, or **no tumor**. Such a tool is intended to support radiologists by providing a fast second opinion, not to replace the doctor's judgement.

The central goal of this work is **not simply to achieve high accuracy**. As explained below, high accuracy on this dataset is already easy and is no longer considered a scientific contribution. Our goal is to build a model that remains accurate even when (a) very few labelled examples are available, and (b) the model is tested in a fair, realistic way. These two properties are what make the work novel and publishable in a good journal.

2. Why a Straightforward Approach Is Not Enough

Before describing our method, it is important to understand two well-documented problems that affect almost every existing study on this dataset. Our approach is specifically designed to solve them.

Problem A — The accuracy is already saturated

Dozens of published papers train a standard model on this dataset and report 98–99% accuracy. Because everyone already reaches this number, simply reporting another high accuracy adds nothing new. A strong journal expects a genuine contribution beyond the headline number.

Problem B — Hidden 'data leakage' inflates the results

An MRI scan of one patient produces many 2-D image slices. If some slices from a patient are used to *train* the model and other slices from the **same** patient are used to *test* it, the model has effectively already seen that patient. The reported accuracy then looks excellent but is misleading — it will not hold up on new patients in a real hospital.

This is called **data leakage**, and published studies have measured that it can falsely raise the reported accuracy by 30–55 percentage points. A correct evaluation must keep all slices from one patient (or one data source) entirely on one side — either training or testing, never both.

An additional data-quality note: this public dataset is assembled from three sources, and one of them is known to contain some mislabelled tumor images. We will clean and document this before training.

Problem C — Labelled medical images are scarce and expensive

Every labelled MRI requires an expert to mark what it shows. In real settings, labelled data is limited while *unlabelled* scans are plentiful. A method that needs only a small number of labels is far more useful and more valuable, both scientifically and commercially.

3. Proposed Approach (Explained Simply)

Our method combines three modern ideas. Each is explained below in plain terms, followed by why we use it.

Building block 1 — A Vision Transformer (the 'eyes' of the model)

A Vision Transformer (ViT) is a type of neural network that analyses an image by cutting it into a grid of small square patches and studying how those patches relate to one another across the whole image. This lets it notice both fine local detail and the overall structure of the scan.

Analogy: an older model reads an image the way someone reads with a magnifying glass — one small region at a time. A Vision Transformer reads it the way an experienced radiologist scans the whole film at once, comparing distant regions to understand the full picture.

Building block 2 — Self-supervised pre-training (learning without answers)

Normally a model only learns from images that already carry the correct answer (the label). Self-supervised learning lets the model first study a large number of MRI images **without any labels at all**. It does this by giving itself little puzzles — for example, hiding part of an image and trying to reconstruct it. By solving millions of such puzzles, the model learns the general visual 'language' of brain MRI: textures, shapes, and what a normal brain looks like.

Analogy: before an exam, a student reads many textbooks to understand the subject in general, long before anyone hands them practice questions with marked answers. Self-supervised learning is that broad reading stage — the model becomes 'fluent' in MRI images before it is ever told which tumor is which.

Building block 3 — Semi-supervised fine-tuning (learning from a few answers)

After the broad self-taught stage, we teach the model the actual four categories using only a **small** set of labelled images, while still drawing on the large pool of unlabelled images to reinforce its understanding. This is the step that delivers our key advantage: strong accuracy from very few labels.

Analogy: because the student has already read widely, they only need to see a handful of worked examples to master the exam — not thousands.

4. Why We Chose This Method

The table below summarises why this combination is the right scientific choice for the goals in Section 1.

Design choice	Reason for choosing it
Vision Transformer backbone	Captures both local detail and whole-image structure; it is the architecture behind today's strongest medical-imaging models.
Self-supervised pre-training	Lets the model learn from abundant unlabelled scans, reducing dependence on expensive expert labels and producing more robust features.
Semi-supervised fine-tuning	Achieves high accuracy using only a small fraction of labels — the central, novel contribution of the study.
Leakage-free patient/source split	Guarantees the reported accuracy is honest and would hold on new patients — the property reviewers and clinicians most demand.
Explainability (visual heat-maps)	Shows *where* the model looks when it decides, so a doctor can trust and verify the output.

5. How We Will Prove the Method Works

To make the claim scientifically convincing, we compare our method against simpler standard approaches under identical, fair conditions. These comparison models are not the product; they exist only as evidence, the way a control group is used in a medical trial.

Model	What it represents
Baseline 1	Standard supervised model (what most existing papers do).
Baseline 2	Same model trained from scratch — tests the value of prior knowledge.
Baseline 3 & 4	Self-supervised model with standard fine-tuning — isolates each component's effect.
Proposed method	Self-supervised + semi-supervised — our complete approach.

Every model is tested across different amounts of labelled data (for example 1%, 5%, 10%, and 100% of labels) and under both the unfair (leaky) split and the fair (patient/source) split. The key results we will report are: (i) how accuracy holds up as labels are reduced, and (ii) how much the unfair split exaggerates accuracy compared with the fair one. These two findings form the backbone of the paper.

6. Expected Contribution and Significance

- **Scientific:** a demonstration that strong brain-tumor classification is possible with only a small fraction of labelled scans, evaluated without data leakage.
- **Methodological honesty:** a clear measurement of how much common evaluation mistakes inflate accuracy — useful to the whole research community.
- **Practical / translational:** a label-efficient, explainable design that is a realistic foundation for a clinical decision-support tool.

Important framing: this system is intended as *decision support for radiologists*, not as an autonomous diagnostic device. Any real clinical use would require separate prospective validation and regulatory approval. This positioning keeps the work both ethically sound and practically deployable.

Prepared as a concise concept note for supervisor review. A detailed methodology, experimental protocol, and implementation plan will follow once the approach is approved.